

Project Scout

Master Project Summary

Last updated: June 9, 2026 | Version 21

Upload this document at the start of every new Claude or ChatGPT conversation about Scout. This is the single source of truth.

June 9, 2026 — What Changed Since June 8:

- Flexible memory foundation completed — Scout can now learn and recall ANY fact, not just 4 hardcoded ones.
- FactKey.kt updated — keeps original 4 fixed keys, adds FactKey.custom() helper for any label.
- TeachExtractor.kt widened — now catches general patterns: 'my favorite ___ is ___', 'my ___ is named ___'.
- RECALL_FACT intent added to IntentType enum in MainActivity.kt.
- handleRecallIntent() added to MainActivity.kt — looks up any fact from TruthDb by label.
- RECALL_FACT routing added to ScoutIntentRouter.kt — catches 'what is my / what's my / do you know my'.
- Tested and confirmed: Scout remembered 'favorite color = teal' and a family member's name with no hardcoding.
- Flexible Memory Planning Document created (Scout_Flexible_Memory_Plan.pdf) — reviewed and approved by both Claude and ChatGPT.

1. Who We Are

Patrick Lippy — developer, project owner, creator of Scout. Not a professional programmer. Stroke survivor, blind in right eye, type 1 diabetic, dyslexic. Explain things calmly and at screenshot level.

- Diana: Patrick's wife
- Elijah: Patrick's son, age 9. Scout's biggest fan. Has drawn pictures of Scout.
- Nicolas: The family dog. The Nicolas Protocol: Scout stops immediately when the dog is detected.

Names must NEVER be hardcoded in Scout's code. Always use variables.

AI Collaborators: Patrick works with both Claude and ChatGPT as project partners. Cross-review between the two is welcome and encouraged. Grok was tried and discontinued.

2. What Scout Is

Scout is a calm family companion robot running on a Samsung Galaxy phone mounted in landscape mode as a permanent face display. Scout has animated eyes, speaks, listens, sees via camera, and remembers the family.

- Package: com.example.scoutface | Language: Kotlin + C++ NDK
- Primary device: Samsung Galaxy A32 — stress-test device, not the benchmark
- Dev device: Samsung Galaxy Fold 7 (wireless via WiFi from Android Studio)
- Future hardware: KEYESTUDIO Mini Tank Kit V2 chassis via Bluetooth (opt-in)
- Ship target: Google Play Store — \$9.99 one-time purchase. No subscriptions. Ever.

3. Identity & Purpose

Scout should feel: Calm. Thoughtful. Emotionally subtle. Grounded. Occasionally curious. Sometimes unsure. Quietly alive. Honest. Predictable. Present.

Scout should NOT: Constantly praise the user. Act overly excited. Feel fake or scripted. Behave unpredictably. Constantly force conversation. Use permanent goodbye language.

Core Philosophy: Stability > Features | Presence > Intelligence | Honest > Fake cheerful | Predictable > Flashy | Local-first > Cloud dependence

4. Business Model

App: \$9.99 one-time purchase on Google Play Store. No recurring fees. No subscriptions. Ever.

Baseline Brain: TinyLlama 1.1B Chat Q4_K_M (~669 MB) ships with every Scout via Google Play Asset Delivery. Runs entirely offline. VERIFIED on BOTH Galaxy A32 and Fold 7.

Optional Gemini: Users add their own free Gemini key in Settings. Scout NEVER ships with a bundled key.

5. Support Scout (In-App, Optional)

Tier	Amount	Label
Coffee	\$3	Buy Scout a Coffee
More	\$5	Support Scout More
Grow	\$10	Help Scout Grow
Founding	\$25	Founding Supporter

\$25 Founding Supporter tier stays — for the person who connects deeply with Scout and wants to be part of the story.

6. Architecture — Five-Layer Memory Stack

Layer	Type	Storage	Status
Working	Sensory	RAM	Done
Habit	Patterns	JSON 14-day	Done
Truth	Authority	SQLite	Done — NOW FLEXIBLE
Relevance	Index	Local Vector	Not yet
Reflective	Wisdom	LLM read-only	Not yet

- Sovereign Rules: SQLite Truth always overrules everything.
- Nicolas Protocol: dog detection → immediate stop.
- Guest Mode: unknown face → 'Hello, I am Scout. What is your name?'
- Flexible Memory: Scout now stores and recalls ANY fact — not just hardcoded family slots.

7. Current Technical State

Working:

- ✓ Animated face (ScoutFaceView) — mouth, iris drift, thinking expression
- ✓ Speech recognition (Android STT) + Text to Speech (TTS)
- ✓ Camera — face detection (ML Kit), scene labeling
- ✓ Gemini API integration (gemini-3.5-flash)
- ✓ Memory layers: TruthDb, ConversationDb, HabitLayer, PeopleDb, JournalDb
- ✓ Intent router — time, date, greetings, family facts, downloads, vision, weather, IDENTITY, RECALL_FACT
- ✓ Teaching system — flexible: 'my favorite color is teal', 'my sister's name is...' → stored permanently
- ✓ Recall system — flexible: 'what is my favorite color?' → Scout reads from TruthDb (NEW June 9)
- ✓ Weather — current, tonight, tomorrow, specific day, 7-day, precipitation % via Open-Meteo
- ✓ ScoutPresenceDecider — four time-of-day modes
- ✓ OFFLINE BRAIN COMPLETE — TinyLlama 1.1B on BOTH A32 and Fold 7
- ✓ Identity questions hardcoded (June 5); routing expanded (June 7)
- ✓ Total offline mode — 'go offline' blocks ALL internet features
- ✓ Thinking-state expression — drift, narrowed lids, asymmetric brows (June 8)

Pending — Do This Next:

- Proposal Sandbox 'Want me to remember that?' confirm step before saving (Step 3 of memory plan).
- Permanent vs temporary memory sorting — birthday vs appointment (Step 4 of memory plan).
- Caring follow-up loop — 'How was your appointment?' then forget (Step 5 of memory plan).
- ScoutFaceView — inner brow knit while thinking (parked June 8). ScoutFaceView cleanup (2 dead-code lines).
- STT phonetic/fuzzy matching — 'Scout' misheard as 'Gal', 'Scott', 'Out'.
- Downloads should respect 'go offline' (finish total-offline mode).

- Response cleanup layer — post-TinyLlama filter before TTS.
- Scout news feed — FUTURE feature, same clean pattern as weather.
- MainActivity cleanup and shortening — ongoing.
- Offline Brain Phase 3 — Google Play Asset Delivery.
- Wire in full mood system (CALM/CURIOUS/HAPPY/THINKING/CONCERNED).

7b. b8946 API Discoveries — CRITICAL

llama_vocab is a separate type: llama_tokenize and llama_token_to_piece now require const llama_vocab*. Get via llama_model_get_vocab(model).

Functions that crash — hardcoded instead: llama_n_vocab → 32000. llama_token_eos → 2. llama_token_eot → 2.

KV Cache: llama_kv_cache_clear() does not exist. Free and recreate context each call. Adds ~300ms but guarantees clean state.

Logits: After prefill, use n_prompt-1. In generation loop, use index 0.

Backend: ggml_backend_load_all() returns 0 on Android. Use ggml_backend_load(cpuPath) with explicit .so path.

Struct padding: _pad[508] trailing array on params structs. Do NOT remove.

Tuning (A32): n_threads=2, n_predict=64, history=2 turns. ~15 tok/s prefill, ~4 tok/s generation. 20–40s per answer — acceptable, Gemini is fast path.

7c. Known Issues — Do Not Touch Without Discussion

Iris jitter (A32 idle): Hardware timing issue. NOT present on Fold 7. Separate from thinking drift (fixed June 8).

Barge-in: Deliberately disabled. Runaway loop. Status: PARKED.

STT name recognition: 'Scout' misheard as 'Gal', 'Scott', 'Out'. Phonetic matching needed.

Live news: Neither brain reads live news. Future news feed needed.

ScoutFaceView dead code: Line 1023 has a doubled condition. Line 709 has unused browAsym. Harmless but messy — clean up in a dedicated session.

7d. Session Log

June 5: IDENTITY intent + hardcoded responses. Weather offline fix. Total offline mode. Offline knowledge confirmed.

June 7: TinyLlama A32 crash stabilized. Identity routing expanded. Face direction locked.

June 8: Thinking expression — gentle iris drift, narrowed lids, asymmetric brows. Flexible Memory Planning Document created and approved by Claude and ChatGPT.

June 9: Flexible memory foundation built. FactKey widened. TeachExtractor generalized. RECALL_FACT intent added. Scout successfully recalled 'favorite color = teal' and a family member's name with zero hardcoding. First true autonomous memory recall.

8. Working Rules — Always Apply

- Full paste-ready replacements only, one file at a time. No snippets. No partial files.
- Surgical CTRL-F and CTRL-R approved for large files — always specify which tab first.
- Some Scout files have NO indentation (flush-left). If a search fails, try a shorter unique single-line string.
- Some logic lives in TWO places (e.g. updateLife AND scheduleNextFrame) — change both or Scout flickers.

- One safe change at a time. Build and test before the next change.
- Never touch speech, camera, or download systems without explicit discussion.
- Never touch ScoutFaceView casually — it is Scout's visual heart.
- Both Claude and ChatGPT are active collaborators — cross-review welcome.
- Patrick is not a professional programmer — explain at screenshot level always.
- Patrick is dyslexic, stroke survivor, blind in right eye, T1 diabetic — keep messages clear and concise.

9. Key Files

File	Description
MainActivity.kt	Main app — all logic. IntentType enum, handleIdentityIntent, handleRecallIntent.
ScoutFaceView.kt	Custom face canvas — all visual animation. Thinking expression updated June 8.
ScoutIntentRouter.kt	Intent routing — IDENTITY + RECALL_FACT added.
TeachExtractor.kt	Extracts facts from speech — NOW FLEXIBLE (any label, any value).
FactKey.kt	Fact labels — fixed keys kept + FactKey.custom() for any new label.
TruthDb.kt	SQLite fact store — was already fully flexible. No changes needed.
GeminiClient.kt	Gemini HTTP wrapper with cooldown discipline
ScoutPromptBuilder.kt	Builds Gemini system instruction
ScoutGeminiManager.kt	Gemini orchestration
ScoutWeatherManager.kt	Live weather via Open-Meteo — precip %, offline-aware
ScoutPresenceDecider.kt	Social timing layer
LlamaEngine.kt	Offline brain JNI wrapper — WORKING
OfflinePromptBuilder.kt	TinyLlama prompt formatter — needs update per brain model
scout_llama_jni.cpp	C++ JNI bridge — compiled into libscout_llama.so
scout_llama_api.h	Self-contained b8946 declarations
CMakeLists.txt	NDK build config
HabitLayer.kt	Pattern memory — 14-day decay

10. Scout Animation Goal & Mood System

Scout's face should feel alive and emotionally present, but always calm. Never perfectly still. Target: subtle organic presence — calm, stable, observant, quietly alive.

- CALM — Eyes center, brows neutral, subtle smile (0.15). Scout's default.
- CURIOUS — Eyes shift right and up, one brow lifts (0.18), slight tilt (0.12).
- HAPPY — Eyes center, both brows lift gently (0.12), smile increases (0.35).
- THINKING — Eyes drift up/around, lids narrow, brows asymmetric. PARTLY BUILT June 8.
- CONCERNED — Eyes look slightly down, inner brows lift (0.05), tilt inward (-0.18).

Face Evolution Roadmap (Locked June 7): v1.0 stable eyes → v1.5 eyebrows + blink → v2.0 simple lids → v3.0 full emotion system. Keep forever: blue iris, white sclera, cartoon style. Never add: tear ducts, skin folds, eyelashes, realistic anatomy. Design principle: 'Scout stays Scout. He just gets a little more alive.'

11. TinyLlama Safety Architecture

Layer 1: Hard-coded safety rules (app level). Layer 2: Scout identity rules (system prompt + hardcoded intents). Layer 3: TinyLlama generation (last line of defense only). TinyLlama knows profanity and scary content — Scout must NOT rely on it for safety decisions. Future: response cleanup layer after TinyLlama, before TTS.

12. Device Requirements

Minimum: Android 13+, 4 GB RAM, ~3–5 GB free. Everything works; responses slower.

Recommended: Android 13+, 8 GB+ RAM, modern CPU (2023+), 10 GB+ free.

Play Store wording: 'Offline AI performance varies by device. Scout runs on many Android devices, but newer phones and tablets provide faster response times and smoother animations.'

13. Brain Upgrade Models

All models already downloaded to: Internal Storage > Scout > Downloaded_Offline_Batch > Future_Upgrades.

Tier	Size	Model
FREE	669 MB	TinyLlama 1.1B Q4_K_M — baseline, always included
\$2.99	1.79 GB	Phi-2 2.7B Q4_K_M — Microsoft. Solid first upgrade.
\$4.99	2.02 GB	Llama 3.2 3B Q4_K_M — Meta. THE most important upgrade.
\$6.99	~2.4 GB	Phi-4 Mini Q4_K_M — Microsoft (PRIMARY Pack 3).
\$6.99 ALT	2.39 GB	Phi-3.1 Mini 4K Q4_K_M — tested backup Pack 3.
\$9.99	4.92 GB	Llama 3.1 8B Q4_K_M — Meta. Flagship, Fold 7 class.

14. Hardware Direction — Optional

Scout works fully without hardware. KEYESTUDIO Mini Tank Kit V2 (Patrick owns one). Hardware Mode is opt-in. The \$9.99 baseline must work fully on the phone alone.

15. Future Vision

Scout notices patterns, generates a suggestion for a new behavior, and asks permission before activating it. Nothing changes without explicit consent. Scout cannot write compiled Kotlin — but CAN generate behavioral scripts, response patterns, routing rules, and habit triggers within the existing safe framework.

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